

# VentureMiner AI

## Report Generation

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# Document 17 — Report Generation

*The report is the artifact. It is what the user ships to a co-founder, a board, an LP, or a client. This document defines the report types, the assembly pipeline, the rendering, and the export formats.*

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## 1. Purpose & Scope

This document is the contract for report generation — the system that consumes evidence, scores, and templates and produces a polished, exportable, citable report.

## 2. Report types

| One-page brief    | 1 page      | Quick share, triage         | < 60s p75    |
|-------------------|-------------|-----------------------------|--------------|
| Full report       | 10–25 pages | Default validation artifact | < 8 min p75  |
| Comparison report | 5–10 pages  | Side-by-side opportunities  | < 3 min p75  |
| Board pack        | 15–30 pages | Quarterly review            | < 15 min p75 |
| Deal memo         | 2–3 pages   | Investor deal flow          | < 90s p75    |

## 3. Assembly pipeline

plan → outline → section drafts (parallel) → verifier pass → assembly → chart render → export

- Plan: chooses sections based on report type, depth, persona.

- Verifier pass: AGT-VERIFY audits each section.
- Assembly: report is composed in order; transitions added.
- Chart render: charts (market size, score breakdown, comparison matrix) are generated from structured data.
- Export: PDF, DOCX, MD, HTML, plus optional PPTX for board pack.

## 4. Section templates

Each report type has a template. The template defines:

- Section order.
- Target length per section.
- Required elements (citations, charts, callouts).
- Style (executive tone vs. analyst tone).

Templates are versioned and editable per workspace (post-MVP).

## 5. Citation handling

- Citations are inline (numbered) and collected in a footnotes section.
- Each citation links to the underlying chunk and source URL.
- Each section shows a freshness class.
- A "View sources" panel lists every cited source with freshness.

## 6. Charts and tables

Charts are generated from structured data, not from LLM output:

- Market size bar (TAM/SAM/SOM).
- Growth line (YoY).
- Demand bars (search/social volumes).
- Score radar (dimension breakdown).
- Comparison matrix (side-by-side).
- Pricing table (competitors, tiers).

All charts include alt text and a data table for accessibility.

## 7. Branding (white-label)

For Team+ (and required for Enterprise):

- Logo (top + bottom of every page).
- Palette (primary, accent).
- Footer (custom text, e.g. "Confidential — Acme Strategy").

Branding is per-workspace; never per-report (to avoid accidental leakage).

## 8. Export formats

| PDF      | reportlab + weasyprint | Default, prints well      |
|----------|------------------------|---------------------------|
| DOCX     | python-docx            | Editable in Word          |
| Markdown | native                 | Lightweight, git-friendly |
| HTML     | native                 | Web sharing               |
| PPTX     | python-pptx            | Board pack (post-MVP)     |

All formats share the same source-of-truth structured report.

## 9. Acceptance criteria

- AC-RPT-0001-1: brief generated < 60s p75.
- AC-RPT-0002-1: full report 10–25 pages.
- AC-RPT-0006-1: PDF, DOCX, MD, HTML available.
- AC-RPT-0008-1: white-label applied (Enterprise).
- AC-RPT-0010-1: every claim footnoted.
- AC-RPT-0011-1: freshness shown per section.

## 10. Cost & latency

See Document 08 §9.

## 11. Failure modes

| Section generation fails | Retry; if persistent, mark section "could not be generated" with explanation. |
|--------------------------|-------------------------------------------------------------------------------|
| Verifier rejects         | Re-generate with feedback; up to 2 retries; then surface.                     |
| Chart data missing       | Render with placeholder; mark freshness.                                      |
| Export fails             | Provide alternative format; surface error.                                    |
| User edits report        | Edits are stored as a "manual override" version; original kept.               |

## 12. Appendix

### 12.1 Revision history

|      |            |          |                        |
|------|------------|----------|------------------------|
| v0.5 | 2026-07-20 | Doc Team | All sections drafted   |
| v1.0 | 2026-07-20 | Doc Team | First approved version |

### 12.2 Cross-references

- Application Flow: Document 03 §7.
- Multi-Agent: Document 08.
- Agent Specs: Document 09.
- RAG: Document 10.
- Scoring Engine: Document 16.

*End of Document 17 — Report Generation. The report is the artifact that makes the AI work shippable.*